THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL UNIT TEST SET 1

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The angle subtended by a diameter at any point on the circle is: 1

(a) 120° **(b)** 60° **(c)** 135° **(d)** 90°

2. The area of a rectangle of length 12 cm and breadth 7 cm is (cm^2): 1

(a) 84 **(b)** 85 **(c)** 86 **(d)** 83

3. A fair die is rolled once. The probability of getting an even number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{4}$

4. For the sequence $t_n = 2n + 1$, the value of t_5 is: 1

(a) 13 **(b)** 10 **(c)** 12 **(d)** 11

5. The radius of a circle of diameter 14 cm is (in cm): 1

(a) 8 **(b)** 7 **(c)** 9 **(d)** 6

6. The perimeter of a rectangle of length 14 cm and breadth 9 cm is (cm): 1

(a) 45 **(b)** 47 **(c)** 46 **(d)** 48

7. A fair coin is tossed once. The probability of getting a head is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$

8. The common difference of the AP $4, 7, 10, 13, \ldots$ is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

9. The perpendicular from the centre of a circle to a chord of length 10 cm bisects it. Each half is (in cm): 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

10. The perimeter of a square of side 9 cm is (cm): 1

(a) 35 **(b)** 38 **(c)** 36 **(d)** 37

11. A bag has 2 red, 3 blue and 5 green balls. The probability of drawing a green ball is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

12. The next term of the AP $2, 7, 12, 17, \ldots$ is: 1

(a) 24 **(b)** 23 **(c)** 21 **(d)** 22

13. The angle subtended by an arc at the centre is 50°. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 35° **(b)** 45° **(c)** 15° **(d)** 25°

14. The area of a triangle with base 12 cm and height 8 cm is (cm^2): 1

(a) 49 **(b)** 47 **(c)** 48 **(d)** 50

15. A ticket is drawn from tickets numbered 1 to 10. The probability that the number is a multiple of 2 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{4}$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. An arc of a circle subtends 36° at the centre. Find the angle it subtends at a point on the remaining part of the circle. 2

17. Find the circumference of a circle of radius 21 cm (take $\pi = \frac{22}{7}$). 2

18. A fair die is rolled. Find the probability of getting a prime number. 2

19. Write the first four terms of the sequence $t_n = 3n + 1$. 2

20. $ABCD$ is a cyclic quadrilateral with $\angle B = 72^\circ$. Find $\angle D$. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A chord of a circle of radius 5 cm is at a distance of 3 cm from the centre. Find the length of the chord. 3

22. The area of a parallelogram is 108 cm^2 and its base is 12 cm. Find its height. 3

23. A bag contains 3 red and 7 blue balls. A ball is drawn at random. Find $P(\text{red})$ and $P(\text{not red})$. 3

24. Find the 10th term of the AP with first term 3 and common difference 4. 3

OR

Points A and B lie on a circle with centre O and $\angle AOB = 48^\circ$. A point C lies on the major arc. Find $\angle ACB$ and say why $\angle ADB = \angle ACB$ for another point D on the major arc. 3

25. Find the area of a triangle with sides 3 cm, 4 cm and 5 cm using Heron's formula. 3

OR

A die was thrown 120 times and the number 4 appeared 22 times. Find the experimental probability of getting 4, and of not getting 4. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A chord of length 24 cm is at a distance of 9 cm from the centre of a circle. Find the radius. 5

OR

A rectangular field is 40 m long and 25 m wide. Find its perimeter and area, and the cost of fencing it at ₹12 per metre. 5

27. Two coins are tossed together. Write the sample space and find the probability of 5

(i) two heads,

(ii) at least one head,

(iii) no head.

OR

A sequence is given by $t_1 = 4$ and $t_{n+1} = t_n + 3$. Write the first five terms, say what kind of sequence it is and find t_{10}. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND